

Tejas Yalamanchili

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EDUCATION

Purdue University

Masters of Science, Computer Science (GPA: 3.77 / 4.00)

- **Achievements:** Dean's and Semesters List all Semesters
- **Coursework:** Compilers, Data Structures and Algorithms, Data Mining And Machine Learning, Intro to Artificial Intelligence, Operating Systems, Database Systems, Reinforcement Learning, Computer Vision Programming, Parallel Computing, Deep Learning

Aug 2024 - May 2026

West Lafayette, IN

Purdue University

Bachelors of Science, Computer Science (GPA: 3.87)

- **Achievements:** Deans and Semesters Honors List all semesters and 2 X hackathon winner
- **Coursework:** Operating Systems, Computer Networks, Databases, Systems Programming, Data Structures and Algorithms

Aug 2021 - May 2025

West Lafayette, IN

SKILLS

- **Languages:** Java, Go, Rust, Python, C/C++, C#, ARM Assembly, HTML, CSS, Javascript, SQL, CUDA, Typescript, GraphQL
- **Frameworks:** Asp .Net, Spring Boot, Flask, React, Flutter, numpy, pandas, PyTorch, Tensorflow, Xcode, MySQL
- **Tools and Technologies:** Azure, AWS, Hugging Face, GitHub, DagsHub, Linux, Kubernetes, Docker

EXPERIENCE

Cummins | Graduate Research Assistant

Aug 2025 - May 2026

- Led a collaborative Purdue Data Mine-Cummins project to build and validate predictive-maintenance AI pipelines on real equipment data, using Python, NumPy, and PyTorch for feature engineering and model training; validated performance against actual failure cases and delivered a functional early fault-detection pipeline.
- Designed an AI-driven agent system that used predictive-maintenance model outputs to detect failures, map them to affected components, and trigger automated supplier part-ordering workflows before warranty expiration, improving reliability and reducing at-risk warranty expirations to ~20%.

Software Engineer Intern

May 2024 - May 2026

- Developed a robust engineer-monitoring interface with Spring Boot and integrated five features, including real-time tracking, to improve system reliability and uptime while reducing issue detection and resolution time.
- Deployed RAG and LlamaIndex-based chatbots for technician assistance using Hugging Face libraries, improving troubleshooting workflows with guided repair steps and context-aware support.

Tech Mahindra | Machine Learning Intern

May 2023 - Aug 2023

- Leveraged CodeBERT to consolidate and categorize similar functions in Nissan's automotive codebase into clusters, then built a semantic code-search system that helped developers navigate the organized clusters and increased productivity by 40%.
- Constructed a chatbot using Retrieval-Augmented Generation (RAG) and prompt engineering, enabling customization for enterprise search and delivering context-aware responses, reducing query resolution time by 25%

Providence Global | Software Engineer Intern

Jun 2022 - Aug 2022

- Engineered a .NET Core web API packaged in Docker containers to support seamless migration of keys and settings to Azure App Configuration, reducing deployment time by 30%.
- Optimized the SQL database for client records through data normalization and indexing, reducing query time from 30 seconds to 4 seconds and improving overall database performance.

AUUG | Software Engineer Intern

May 2022 - Jun 2022

- Developed a C# library with .NET to stream musical information to multiple devices, expanding cross-platform availability.
- Implemented the O2 communication protocol in C# using asynchronous sockets to enable efficient data transmission and improve the user experience.
- Optimized the application's performance, achieving a 15% reduction in latency and improving the overall responsiveness and reliability of the streaming service

PROJECTS

AgentTrail | <https://agent-relay-website.vercel.app/>

Jan 2026 - Present

- Built and deployed a Python-based, transaction-aware agent runtime library, instrumenting multi-agent workflows with structured run logging and API key management to make failures traceable and reproducible across deployments.
- Developed and deployed a scalable web platform using React, FastAPI, and AWS services including S3, CloudFront CDN, and Elastic Beanstalk, providing real-time dashboards, OAuth-secured access, and API-driven agent monitoring.

Fit Check

Jan 2024 - Mar 2024

- Built an e-commerce platform aggregating apparel from over 10 leading brands and launched a virtual try-on feature using a Stable Diffusion model with enhanced image preprocessing, including precise segmentation and identification of critical fit points, increasing cloth-mapping accuracy on sample avatars by 40%.
- Built the full-stack application using Django on the backend and a TypeScript-based frontend, implemented Retrieval-Augmented Generation for dynamic recommendation prompts, and optimized application performance by reducing page-load latency.

UniLeap

May 2023 - Dec 2023

- Co-founded UniLeap to provide a personalized mentoring platform for high schoolers and graduate students needing college application and visa assistance for studying abroad
- Steered the deployment of Django backend on AWS EC2, and used Nginx as a load balancer and React.js frontend on AWS Amplify to provide optimal performance and a user-friendly experience for 10+ active students.